THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 8: **Water Resources (Geography)** | Class 10 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** A is true but R is false.	A true; R false
2	**(b)** reduced storage capacity	Sediment fills the reservoir
3	**(a)** Both A and R are true and R is the correct explanation of A.	Both true; R explains A
4	**(d)** Tehri dam	In Uttarakhand
5	**(c)** rainwater harvesting	Harvesting adds supply
6	**(d)** A is false but R is true.	A false; R true
7	**(a)** local rainwater harvesting	Locally suited
8	**(b)** water is a limited resource	The usable share is small
9	**(a)** Both A and R are true and R is the correct explanation of A.	This integration was the reason he gave for the statement; R explains A.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true statements about different multi-purpose projects; R does not explain A.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	They change river flow and trap silt (1); They block fish migration and flood forests (1).	1+1
12	Water is scarce (1); Efficient methods reduce wastage and make water available for more users (1).	1+1
13	Crops need regular water (1); Scarcity lowers yields and income, and forces farmers to deepen wells (1).	1+1
14	Silt reduces storage so floods spill over (1); Sudden release of water from full dams floods downstream areas (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	People need water as a right (1); Disputes arise over who controls it (1); Fair access should be ensured for all (1).	1+1+1
16	The law made rooftop harvesting compulsory (1); Strict enforcement and awareness helped (1); Groundwater levels rose in Chennai and other places (1).	1+1+1
17	Tankas store water in dry areas (1); Kuls use mountain streams (1); Inundation channels use floods (1).	1+1+1
	OR: Benefits of dams reach some regions only (1); Displaced people may not gain (1); Inequalities between regions can grow (1).	1+1+1
18	It uses local bamboo and spring water (1); Water is delivered drop by drop (1); It needs no energy and wastes little (1).	1+1+1
	OR: High-yielding seeds need more irrigation (1); Farmers use tube wells (1); Water tables fell (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) A project serving many purposes together such as irrigation, power generation, flood control and water supply (1). (ii) Mahanadi (1). (iii) Hydel power production or irrigation (1). (iv) Damming affects natural flow, causes excessive sedimentation in the reservoir and fragments rivers, harming aquatic life; it also displaces local communities and can increase floods (2).	1+1+1+2
20	Sringaverapura near Allahabad had a sophisticated system to channel the floodwaters of the Ganga (1); During Chandragupta Maurya's time, dams, lakes and irrigation systems were built extensively (1); Evidence of irrigation works has been found in Kalinga (Odisha), Nagarjunakonda (Andhra Pradesh), Bennur (Karnataka) and Kolhapur (Maharashtra) (1); In the eleventh century, Bhopal Lake was one of the largest artificial lakes of its time (1); In the fourteenth century, the tank at Hauz Khas in Delhi was built to supply water to the Siri Fort area (1).	1+1+1+1+1

	OR: Western Himalayas use guls or kuls, channels that carry stream water to terraced fields (1); In the arid and semi-arid Rajasthan, rooftop rainwater is stored in underground tanks (tankas) in Bikaner, Phalodi and Barmer, and khadins and johads hold monsoon run-off on farms (1); In Bengal's flood plains inundation channels were used to carry floodwater to the fields (1); In Meghalaya, a 200-year-old bamboo drip irrigation system carries spring water to plants (1); Each method uses local material and local rainfall and terrain, which is why it suits its region (1).	1+1+1+1+1

Mistakes to avoid:

- Saying water scarcity is caused only by **low rainfall**; over-use, unequal access, pollution and population growth also cause it.
- Matching projects to the wrong rivers: **Hirakud** (Mahanadi), **Bhakra-Nangal** (Sutlej-Beas), **Sardar Sarovar** (Narmada).
- Writing that Mahatma Gandhi called dams the 'temples of modern India'; the phrase is **Jawaharlal Nehru's**.
- Describing multi-purpose projects as only for **irrigation**; they also give power, flood control, water supply and navigation.
- Mixing traditional systems: **tankas** (Rajasthan), **kuls/guls** (Western Himalayas), **bamboo drip** (Meghalaya), **inundation channels** (Bengal flood plains).
- Forgetting that dams have **disadvantages** as well: sedimentation, displacement and damage to aquatic habitats.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 8: **Water Resources (Geography)** | Class 10 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** submerge forests and affect fish and river flow	Environmental cost
2	**(a)** concrete surfaces prevent percolation	Paved surfaces
3	**(d)** rainwater harvesting	Helps percolation
4	**(a)** Both A and R are true and R is the correct explanation of A.	Both true; R explains A
5	**(a)** Both A and R are true and R is the correct explanation of A.	Both true; R explains A
6	**(a)** evaporation	Water goes up as vapour
7	**(d)** rooftop rainwater harvesting	State policy
8	**(d)** displacement of people	Resettlement problems
9	**(c)** A is true but R is false.	A is true. The project covers four states (Maharashtra, Madhya Pradesh, Gujarat and Rajasthan), so R is false.
10	**(d)** A is false but R is true.	Regulating and damming affects natural flow, so A is false; R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	They suit local climate and soil (1); They were developed by communities over generations (1).	1+1
12	They submerge villages and displace people (1); Benefits go to some regions while costs fall on others (1).	1+1
13	Urban demand is high and sources are limited (1); Reuse cuts demand and reduces sewage (1).	1+1
14	Government, industry and citizens all use water (1); Each must save and protect it for future generations (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Water scarcity is a long-term shortage due to overuse and unequal distribution (1); Drought is a temporary lack of rain (1); Scarcity can occur even in high rainfall regions (1).	1+1+1
16	Traditional tanks and kuls give local solutions (1); Modern projects give large supply (1); Together they provide sustainable use (1).	1+1+1
17	Cities grow and demand more water (1); Concrete reduces recharge (1); Pollution adds to scarcity (1).	1+1+1
	OR: Local needs differ (1); Communities can manage resources better (1); It avoids large-scale displacement (1).	1+1+1
18	Glaciers melt and rainfall patterns change (1); Floods and droughts increase (1); Water supply becomes uncertain (1).	1+1+1
	OR: Drip and sprinkler irrigation use less water (1); Crop choice can suit local water (1); Mulching and soil conservation save moisture (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Underground tanks for storing drinking water (1). (ii) Rainwater collected from rooftops (1). (iii) Bikaner, Phalodi or Barmer (any one) (1). (iv) Rainfall is scanty and uncertain, groundwater is scarce, so harvesting rain gives a safe local source of water throughout the year and reduces dependence on distant sources (2).	1+1+1+2
20	The Narmada Bachao Andolan opposed the Sardar Sarovar Dam because it would displace tribal and other villagers in Gujarat, Madhya Pradesh and Maharashtra (1); Protests also arose against the Tehri dam on the Bhagirathi in Uttarakhand (1); Participants raised questions about resettlement, loss of livelihood and environmental damage (1); They argued that the benefits go to some while the costs fall on local people (1); These movements forced attention to rehabilitation and encouraged the consideration of smaller, local alternatives (1).	1+1+1+1+1

	OR: The government makes laws, builds and manages projects, regulates groundwater and promotes harvesting, as Tamil Nadu did (1); It can enforce pollution control and fair sharing of river water (1); Citizens can save water, repair leaks and avoid wastage (1); Households can harvest rainwater and reuse water (1); Communities can build and maintain structures like johads and kuls, and the combined effort conserves water (1).	1+1+1+1+1

Mistakes to avoid:

- Saying water scarcity is caused only by **low rainfall**; over-use, unequal access, pollution and population growth also cause it.
- Matching projects to the wrong rivers: **Hirakud** (Mahanadi), **Bhakra-Nangal** (Sutlej-Beas), **Sardar Sarovar** (Narmada).
- Writing that Mahatma Gandhi called dams the 'temples of modern India'; the phrase is **Jawaharlal Nehru's**.
- Describing multi-purpose projects as only for **irrigation**; they also give power, flood control, water supply and navigation.
- Mixing traditional systems: **tankas** (Rajasthan), **kuls/guls** (Western Himalayas), **bamboo drip** (Meghalaya), **inundation channels** (Bengal flood plains).
- Forgetting that dams have **disadvantages** as well: sedimentation, displacement and damage to aquatic habitats.

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MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 8: **Water Resources (Geography)** | Class 10 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** tanka - Assam	Tankas are found in Rajasthan
2	**(a)** Both A and R are true and R is the correct explanation of A.	Both true; R explains A
3	**(d)** tanka	A traditional structure of the Thar
4	**(a)** over-extraction of groundwater	Extraction exceeds recharge
5	**(d)** displacement and sedimentation	Costs of dams
6	**(b)** harvesting and reusing water	Conserves for future
7	**(a)** over-extraction and drought	Falling water tables
8	**(b)** Hirakud - Mahanadi	Hirakud is in Odisha
9	**(a)** Both A and R are true and R is the correct explanation of A.	The water shortage makes households collect rain from rooftops; R explains A.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true, but they are different traditional systems in different regions.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Without trees run-off increases (1); Rivers get silted and dry in summer (1).	1+1
12	Rivers cross borders and each state wants more (1); Historical rights and politics complicate agreements (1).	1+1
13	Cities have concrete surfaces that block recharge (1); Harvesting captures run-off and recharges groundwater (1).	1+1
14	They use local materials and simple designs (1); They need no big machinery or compensation costs (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Dams supply power and water to industry (1); They support urban growth (1); Industry then expands near projects (1).	1+1+1
16	Narmada protest was against Sardar Sarovar in Gujarat (1); Tehri protest was against the dam on the Bhagirathi in Uttarakhand (1); Both raised displacement and environmental issues (1).	1+1+1
17	Effluents pollute rivers and groundwater (1); Polluted water is unfit for use (1); Usable supply falls (1).	1+1+1
	OR: Forests hold rain and recharge groundwater (1); Roots reduce run-off (1); Streams flow longer (1).	1+1+1
18	Dams provide irrigation, power and flood control (1); But they displace people and harm ecology (1); So they are important but their costs must be weighed (1).	1+1+1
	OR: Tribunals decide shares between states (1); Their awards are legally binding but sometimes resisted (1); Political will is also needed (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Hydrological (water) cycle (1). (ii) Population growth, over-exploitation, unequal access or pollution (any one) (1). (iii) Kuls (guls) or bamboo drip irrigation (1). (iv) Rooftop rainwater harvesting; fixing leaking taps; reusing water for gardens (any two) (2).	1+1+1+2
20	Large dams are costly, take years to build, displace people and change river ecology, but supply power and irrigation on a large scale (1); Community systems like johads, tankas and kuls are cheap, quick, use local knowledge and recharge groundwater (1); Large projects cause conflict between states and users, while community systems are managed locally (1); Large dams suffer from sedimentation, whereas small structures are easy to maintain (1); A mixed approach, with small structures for local needs and dams for large demands, is most sustainable (1).	1+1+1+1+1

	OR: Rivers flow through several states, so a dam or extraction in one state affects the downstream state (1); States demand larger shares as demand grows, as in the Krishna-Godavari dispute (1); Disputes between users also arise, such as farmers versus industries (1); They can be resolved through negotiation, tribunals and agreements based on fair sharing (1); Efficient use, conservation and cooperative management reduce conflict (1).	1+1+1+1+1

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